Capstone

Huxley Lewis

November 2025

How constraint satisfiability problems and information theory, along with, philosophies of empiricism and determinism implicate the future of artificially intelligent structures.

1 Abstract

2 Introduction

Artificial Intelligence, or AI, has grown significantly over the last decade with the rise of large language models and other foundation models [1]. These models are examples of neural networks, which learn from data rather than being programmed with exact task-specific behaviours [2]. Scaling studies of language models suggest that performance improves with additional parameters, data, and compute, while also making the cost of training and inference a central practical constraint [3, 4]. Although the exact internal mechanisms by which LLMs generalize across new tasks remain only partially understood [1], <their shared structure of data, processing, and generation can still be used to frame arguments about the behaviour and limits of AI systems.> <This structural view of artificial intelligence motivates the use of constraint satisfiability problems, information theory, epistemology, and determinism when reasoning about the limitations of generalized AI models.>

Epistemology is the study of human knowledge [5]. It has also been used in recent work to analyse AI systems as black boxes and to reason about the limits of explanation, justification, and understanding in machine learning [6, 7]. <Because data is a fundamental element of AI systems, epistemic questions about what is known, how it is learned, and what remains inaccessible apply directly to AI.> <The different methods by which an AI system acquires and represents data also shape its structure, including the contrast between symbolic and connectionist approaches.> (citation)

Constraint Satisfiability Problems, or CSPs, are a general framework for representing problems in terms of variables, domains, and constraints [8]. This abstraction makes CSPs useful for discussing classes of problems independently of the details of any particular solver [9]. CSPs have long been used in symbolic

AI to represent relational structure in domain-specific problems while supporting general solving procedures [8, 9]. <This style of representation supports mathematical arguments that may extend to AI systems when those systems can be modeled in CSP-like terms.>

Information theory is a framework for quantifying information, communication, and uncertainty [10]. In machine learning, information-theoretic methods have been used to study how representations preserve, compress, and transform information during learning [11, 12]. Information theory therefore offers one way to analyse the internal behaviour of contemporary connectionist models [12]. <Determinism is the view that events are fixed by prior conditions rather than arising independently.> (citation) <Although this remains debated across philosophy and physics, it provides a framework for asking whether future states of a system are in principle knowable from present information.> (citation)

3 Literature Review

Turing describes the most general computer when he introduces the *a* machine [13] it utilizes a tape as a way to maintain its memory, which in this case describes the computational system's ability to store information. <In other computational systems, memory might describe several structures varying in complexity, but that can all be understood to refer to an information store.> (citation) <Turing machines utilize very simple instructions that can be used to form algorithms, which can describe any kind of operation that a computer is capable of.> (citation) Algorithms are in this case manipulators of information [14], turning it from one form into another. There are many ways of formalizing this kind of information processing, the major being predicate logic, equivalent to lambda calculus, and an extension of predicate logic, constraint satisfiability problems, where predicates are given globalized variables informing [14]. <CSPs in this way can represent many computational problems and are important to describing algorithmic intelligence.> (citation) <These systems have a mathematical formalization as a triple of Variables, Domains, and Constraints.> (citation)

They are able to easily represent many kinds of problems in fields such as logistics, and planning in ways that computers can attempt to solve [14]. <CSPs have been studied for properties of tractability and satisfiability, since knowing this kind of information can support heuristics when solving these industrial problems.> (citation) <These limitations are also important when discussing other applications of CSPs.> <In AI, CSPs are able to describe many kinds of structures, and therefore, can infer limitations of AI models.>

In many works that influenced many popular interpretations of AI, writers such as Isaac Asimov [15] discussed the goals that many hold for AI models. <That AI should mimic the behaviour of humans is one influential perspective, but it also carries important conceptual flaws.> <These thought experiments provide one basis for understanding both the AI of today and a possible AI of the future.> <Humans have several flaws of logic and reasoning that may not be overcome when we aim to create models that mimic ourselves.> (citation) The Chinese-room experiment [16] is another example of a thought experiment which asks us to question the ability of computers to truly understand. <The ability for a machine to give the right output gives no indication of whether the machine understands the input.> (citation) <This has many interpretations, some referring to the lack of true artificial intelligence, while others refer to limitations of the input.> (citation) <The latter applies most in this circumstance, with people being the inspiration for AI architecture, they will be forever limited by the behaviours of people.>

The current theories of knowledge in people originate from the debate of empiricists and rationalists [17]. <These are opposing schools of thought.> (citation) <The difference between them is often framed as one between pre-ordained knowledge that is discovered and empirical knowledge that is gained from experience.> (citation) <Both of these interpretations have implications for limits of human knowledge.> (citation) <Empiricists require that thought be testable and grounded in reality.> (citation) <Rationalists hold that knowledge is in some sense already predisposed, and that we unlock access to it.> (citation) <Regardless of the mechanism of knowledge acquisition, there is information that is unattainable.> <As shown in thought experiments such as Newcomb's problem [18], future events are often treated as examples of the unknowable.>

4 The Model

<The model I have constructed so-far contains three axioms:>

1. <Generation from information is impossible if there exists no relationship between the learned data and generated data.>
2. <Generation from information is impossible if there exists no algorithm to represent the relationship that can be computed in finite time and space.>
3. <Generation from information is equivalent, while the information is the same.>

<This implies three principles,>

i. <Dependency.>

- <From (1). Generation is dependent on the relationships, not on the objects.>

ii. <Generation Uncertainty.>

- <From (2). The more accurate a prediction, the less computable it becomes.>

iii. <Similarity.>

- <From (3). Any relationship can be modelled with an equivalent set of relations.>

<These principles can be used to inform predictions. That is,>

1. <For all AI models, there exists an equivalent Constraint Satisfaction Problem.>
2. <For all AI models, the relationship between generation and information is proportionately tractable to the similarity between the information and the generation.>
3. <For all AI models, there exists no equivalent process to generating the future to absolute detail at the current speed of time that is tractable.>

References

[1] Rishi Bommasani, Drew A. Hudson, Ehsan Adeli, Russ Altman, Simran Arora, Sydney von Arx, Michael S. Bernstein, Jeannette Bohg, Antoine Bosselut, Emma Brunskill, et al. On the opportunities and risks of foundation models. *arXiv preprint arXiv:2108.07258*, 2021.

[2] Yann LeCun, Yoshua Bengio, and Geoffrey Hinton. Deep learning. *Nature*, 521(7553):436–444, 2015.

[3] Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*, 2020.

[4] Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, et al. Training compute-optimal large language models. *arXiv preprint arXiv:2203.15556*, 2022.

[5] Matthias Steup, Ram Neta, and John Turri. Epistemology. In Edward N. Zalta and Uri Nodelman, editors, *The Stanford Encyclopedia of Philosophy*. Metaphysics Research Lab, Stanford University, Fall 2024 edition, 2024.

[6] Manuel Carabantes. Black-box artificial intelligence: An epistemological and critical analysis. *AI & Society*, 35(4):943–952, 2020.

[7] Florian Funer. Accuracy and interpretability: Struggling with the epistemic foundations of machine learning-generated medical information. *Philosophy & Technology*, 35(2):1–28, 2022.

[8] Alan K. Mackworth. Consistency in networks of relations. *Artificial Intelligence*, 8(1):99–118, 1977.

[9] Rina Dechter. Backtracking algorithms for constraint satisfaction problems: A survey. *Computational Intelligence*, 4(3):273–289, 1988.

[10] Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27(3):379–423, 1948.

[11] Naftali Tishby and Noga Zaslavsky. Deep learning and the information bottleneck principle. *2015 IEEE Information Theory Workshop (ITW)*, pages 1–5, 2015.

[12] Ravid Shwartz-Ziv and Naftali Tishby. Opening the black box of deep neural networks via information. *arXiv preprint arXiv:1703.00810*, 2017.

[13] Alan Mathison Turing et al. On computable numbers, with an application to the entscheidungsproblem. *J. of Math*, 58(345-363):5, 1936.

[14] Stuart Russell, Peter Norvig, and Artificial Intelligence. A modern approach. *Artificial Intelligence. Prentice-Hall, Egnlewood Cliffs*, 25(27), 1995.

[15] Isaac Asimov. *I, Robot.* Gnome Press, New York, 1950. A collection of nine science fiction short stories.

[16] Klaus K Obermeier. Wittgenstein on language and artificial intelligence: The chinese-room thought experiment revisited. *Synthese*, pages 339–349, 1983.

[17] Peter Markie and M. Folescu. Rationalism vs. Empiricism. In Edward N. Zalta and Uri Nodelman, editors, *The Stanford Encyclopedia of Philosophy.* Metaphysics Research Lab, Stanford University, Spring 2023 edition, 2023.

[18] Robert Nozick. Newcomb's problem and two principles of choice. In *Essays in honor of carl g. hempel: A tribute on the occasion of his sixty-fifth birthday*, pages 114–146. Springer, 1969.